

HearSense: Intelligent Sinhala Sign Language Communication Platform with Emotion Recognition and Environmental Awareness

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Abstract—Communication accessibility remains a critical concern for individuals who are deaf, hard of hearing, or have speech impairments, particularly in regions where sign language resources and awareness are limited. In Sri Lanka, the absence of integrated tools that support real-time interaction, learning, and environmental awareness further intensifies these challenges. This paper presents a unified intelligent platform designed to enhance Sinhala Sign Language (SSL)-based communication through the integration of multiple assistive technologies. The proposed system combines four key functionalities within a single framework. A multimodal recognition module interprets sign language using hand, facial, and body features to generate real-time Sinhala text. An adaptive learning component personalizes the learning process through performance-based feedback and dynamic difficulty adjustment. An environmental sound detection module utilizes deep learning techniques to identify critical sounds and deliver visual alerts, improving user safety. Additionally, an interactive two-way communication interface enables effective interaction between children with speech disabilities and their caregivers. Evaluation results indicate that the system achieves promising performance across all components, with high accuracy in sound detection, effective sign recognition through multimodal fusion, and reliable prediction in communication support. The integration of these modules enhances usability, engagement, and situational awareness. The proposed platform demonstrates the potential of combining artificial intelligence and human-computer interaction to develop scalable and inclusive communication solutions for underrepresented sign language communities.

Keywords—Sinhala Sign Language, Multimodal Recognition, Adaptive Learning, Environmental Sound Detection, Assistive Communication, Two-Way Interaction, Emotion Recognition

I. INTRODUCTION

Ensuring effective communication for individuals with hearing and speech impairments remains a significant challenge, particularly in contexts where assistive resources are limited. In Sri Lanka, users who rely on Sinhala Sign

Language (SSL) often encounter barriers when interacting with others due to the lack of widely available communication tools and limited public familiarity with SSL. These challenges extend beyond communication alone, affecting learning opportunities, social participation, and overall quality of life.

Recent advancements in artificial intelligence have led to the development of various assistive systems aimed at improving accessibility [3]. These include sign language recognition models, adaptive learning platforms, and alternative communication interfaces. While such systems have shown promising results, they are typically designed to address specific tasks in isolation. As a result, users are required to rely on multiple disconnected tools, which limits usability and reduces the effectiveness of assistive support in real-world scenarios. Furthermore, many existing solutions lack the ability to incorporate contextual information, such as emotional expression or environmental awareness, which are essential for natural and meaningful communication.

To address these limitations, this research introduces a unified intelligent platform that integrates multiple assistive functionalities tailored to Sinhala Sign Language users. The system is designed to support both communication and learning through four interconnected components. These include a multimodal sign language recognition module for real-time interpretation, an adaptive learning system that personalizes training experiences, an environmental sound detection module that enhances user safety, and an interactive two-way communication interface for children with speech disabilities and their caregivers.

By combining these components into a single framework, the proposed system aims to provide a more cohesive and user-friendly solution compared to existing approaches. The integration of multimodal recognition, adaptive learning, and environmental awareness enables a richer and more context-aware interaction experience. Ultimately, this work

contributes to the development of inclusive assistive technologies that support effective communication and independent living for individuals with hearing and speech impairments.

II. LITERATURE REVIEW

The field of assistive technology has seen rapid growth in recent years, with increasing emphasis on improving accessibility and independence for individuals with hearing and speech impairments. Among these developments, sign language recognition systems, intelligent tutoring platforms, environmental awareness tools, and interactive communication applications have emerged as key research directions. Despite this progress, existing solutions often operate as standalone systems, addressing specific challenges without considering the broader communication needs of users in real-world contexts.

Sign language recognition has been widely explored using computer vision and deep learning techniques. Early approaches primarily relied on image-based classification of hand gestures, while more recent studies have incorporated temporal modeling to capture dynamic movements. However, many of these systems are limited by their focus on manual features alone, overlooking the importance of non-manual cues such as facial expressions and body posture. Since sign language is inherently multimodal, the absence of these elements can lead to incomplete or inaccurate interpretation, particularly in complex conversational scenarios.

In parallel, adaptive learning systems have been introduced to enhance the process of acquiring sign language skills[4]. These systems aim to personalize the learning experience by adjusting content based on user performance [1]. While some approaches utilize machine learning techniques to provide feedback, the integration of continuous performance tracking and dynamic curriculum adaptation remains relatively limited. As a result, many platforms still follow semi-static learning paths, which may not effectively support long-term retention or learner engagement.

Another important area of research involves environmental sound recognition, which plays a critical role in improving situational awareness for hearing-impaired individuals. Existing systems typically employ audio feature extraction methods, such as Mel-Frequency Cepstral Coefficients (MFCCs), combined with classification models to identify common environmental sounds[10],[13]. However, challenges such as background noise interference, delayed response times, and lack of prioritization mechanisms can reduce the practical effectiveness of these systems in real-life environments.

Furthermore, communication tools designed for individuals with speech impairments often focus on augmentative and alternative communication (AAC) methods, including symbol-based interfaces and text-to-speech systems. While these tools provide basic support, they frequently lack interactive capabilities and may not be intuitive for young users. The absence of adaptive and predictive features further limits their ability to facilitate natural and efficient communication between users and caregivers.

Given these limitations, there is a growing need for integrated systems that combine multiple assistive functionalities into a single, cohesive platform. A unified

approach that incorporates multimodal sign language recognition, adaptive learning, environmental awareness, and interactive communication can better address the complex and dynamic needs of users. Such integration not only enhances usability but also enables more meaningful and context-aware interactions. This research builds upon these insights by proposing a comprehensive system tailored to the Sinhala Sign Language domain, aiming to bridge existing gaps and improve overall accessibility.

III. METHODOLOGY

HearSense refers to an app that has been designed using a system-based methodology to facilitate learning and communication for those people who are either deaf or have speech impairments. Artificial intelligence technologies such as computer vision and machine learning have been combined with user-centered design principles to create an efficient application. The methodology mainly focuses on the development and integration of four core elements in a system to deliver real-time performance:

- A. Adaptive Sinhala Sign Language Learning System
- B. Environmental Sound Alert Module
- C. Sinhala Sign Language Reader
- D. Sinhala Two-Way Communication System for Children with Speaking Disabilities

A. Adaptive Sinhala Sign Language Learning System

Personalized SSL learning using Reinforcement Learning for dynamic difficulty adaptation, feedback selection, and spaced repetition scheduling based on learner performance [5].

Dataset

In order to construct the dataset, implemented an image collection algorithm in real-time using OpenCV and a single-hand detection module (cvzone HandTrackingModule) to extract the Sinhala sign letters from a camera. The dataset was created class-wise by labeling each sample with a specific number corresponding to a unique Sinhala letter. In each frame, the hand area was extracted by applying a predefined margin around the detected hand position and resized to (300 $\times$ 300) with a white background. This ensured that the aspect ratio was preserved during image processing to prevent any shape deformation. Moreover, the dataset was constructed by saving samples as either correct or wrong according to whether they belonged to the target gesture.

Hand Detection Pipeline

Input frame $\rightarrow$ Detection (cvzone/MediaPipe, confidence ≥ 0.8) $\rightarrow$ Preprocessing (letterbox 640 $\times$ 480 $\rightarrow$ hand crop $\rightarrow$ white canvas 300 $\times$ 300 $\rightarrow$ CLAHE + gamma $\rightarrow$ 224 $\times$ 224) $\rightarrow$ Smoothing (5-frame majority voting) $\rightarrow$ Normalized output

RL Formulation

State Space: 6D vector (accuracy, confidence, response_time, streak, recency, difficulty)

$$|S| = 5 \times 5 \times 4 \times 4 \times 4 \times 4 = 6,400$$

Action Space: 3D vector (difficulty $\times$ feedback_type $\times$ review_interval)

$$|A| = 3 \times 4 \times 3 = 36$$

Reward Function:

$$R(s, a, s') = 0.30R_{acc} + 0.30R_{imp} + 0.25R_{ret} + 0.15R_{eng}$$

Q-Learning Update:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[R + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Key Functionalities

Curriculum: 6 levels (Static: beginner → intermediate → advanced; Dynamic: 3 levels)

Difficulty Adjustment:

- Increase: accuracy $\geq 85\% \wedge$ streak ≥ 3
- Decrease: accuracy $< 60\% \wedge$ attempts ≥ 5

Spaced Repetition (SM-2):

$$I_{next} = 24h \times \begin{cases} 2.5 & \text{if } C \geq 0.80 \\ 1.8 & \text{if } 0.70 \leq C < 0.80 \\ 0.5 & \text{if } C < 0.70 \end{cases}$$

Mastery Levels: 0=New, 1=Learning, 2=Familiar, 3=Mastered

Feedback Q-Learning: Online trained on 12-message library, learns best feedback per state

Implementation Status

Completed: Hand detection pipeline, 6-level curriculum, state/action spaces, Q-agents, SM-2 spaced repetition, mastery tracking, performance persistence, API endpoints

Next: Deep Q-Network, Thompson Sampling, live user testing, A/B testing

Expected Outcome

>15% faster mastery and >80% retention vs. fixed curriculum

B. Environmental Sound Alert Module

The Environmental Sound Alert module detects dangerous environmental sounds in real time using the mobile phone microphone, and notifies deaf and hearing-impaired users through graded visual alerts [11]. Audio is continuously recorded, MFCC features are extracted, and a CNN classifies the surrounding sound. The result is delivered to the React Native frontend via a Flask REST API.

Dataset

Two datasets were combined into a corpus covering seven sound categories: four horn types (bus, car, train, truck) and three siren types (ambulance, firetruck, police) [11]. Traffic ambient noise was excluded as it is continuous and non-directional[12]. Class imbalance was addressed using per-class sample weights during training.

Dataset Statistics:

Category	Sound Classes	Clips	Train	Val	Test
Vehicle Horns	Bus, car, train, truck	~730	467	117	146
Sirens	Ambulance, fire truck, police	~2,500	1,600	400	500
Total	7 classes	~3,230	~1,522	268	448

Fig. 1. Dataset Statistics

Format : WAV / MP3, variable duration and sample rate
Split : 64% train / 16% validation / 20% test (stratified)

Feature Extraction

Audio files were loaded with librosa at 22,050 Hz (mono), truncated or zero-padded to 2.5 seconds, and silence-trimmed before feature extraction.

Mel-Frequency Cepstral Coefficients (MFCCs) were chosen as the feature representation because they model human auditory perception and are robust to variations in recording environment and microphone hardware [10], [13].

MFCC parameters:

Coefficients (n_mfcc) : 13

Time frames (n_frames) : 40

Feature matrix shape : 13 x 40 (520 dimensions total)

Per-coefficient z-score normalization was applied and training-set statistics were saved for identical reapplication at inference. The normalized matrix was reshaped to (13 x 40 x 1) as a single-channel 2D CNN input.

Model Architecture

CNN treating the MFCC matrix as a 2D spectral image[14]:

Input (13 × 40 × 1)

→ Conv2D(32) + BatchNorm + MaxPool(2×2)
+ Dropout(0.25)

→ Conv2D(64) + BatchNorm + MaxPool(2×2)
+ Dropout(0.25)

→ Conv2D(128) + BatchNorm + MaxPool(2×2)
+ Dropout(0.30)

→ Flatten → Dense(256) + Dropout(0.50)
→ Dense(128) + Dropout(0.40)

→ Dense(7, Softmax)

~293,000 parameters · Loss: Sparse Categorical Cross-Entropy · Optimiser: Adam

Fig. 2. Model Architecture

Training

Batch 32 · 100 epochs max · stopped at 76 (EarlyStopping patience 15)

ModelCheckpoint (best val accuracy)

· ReduceLROnPlateau (factor 0.5)

Implementation Status

Completed: MFCC pipeline, CNN training, Flask API, React Native three-tier alert display (full-screen overlay / auto-dismiss banner / silent log), real-time loop.

Next: Alert prioritization (siren > heavy horn > car horn when concurrent detections occur) · TFLite on-device inference · user testing.

Expected Outcome

Real-time detection of 7 sound classes at >90% accuracy with graded alerts deliverable to deaf users, and a ~293K-parameter model suitable for future on-device mobile deployment.

C. Sinhala Sign Language Reader

Real-time multimodal sign recognition using a multi-stream deep learning architecture that independently processes hand gestures, facial expressions, and body posture before fusing them for accurate classification across 383 Sinhala sign vocabulary items.

Dataset

The signVideo dataset is a locally collected Sinhala Sign Language video corpus comprising 8,472 videos across 383 sign classes (approximately 22 videos per class)[9]. The vocabulary spans greetings, numbers, colours, common nouns, verbs, and grammatical particles. A stratified split was applied: 5,827 videos (68.8%) for training, 1,358 (16.0%) for validation, and 1,287 (15.2%) for testing. Labels are stored in a unified label_mapping.json file shared across training and inference. The dataset is significantly under-resourced at approximately 22 videos per class, necessitating targeted strategies to mitigate overfitting.

Feature Extraction Pipeline

Feature extraction uses the Google MediaPipe Tasks API (v0.10.21) [7] with three pre-trained landmark models deployed on GPU (float16): Hand Land marker, Face Land marker, and Pose Land marker. Each video is decoded frame-by-frame via OpenCV, converted to RGB, and processed through a fixed temporal window of 60 frames. Three feature streams are extracted per frame:

- Hand Stream (126 dims) — Both hands tracked simultaneously; 21 landmarks $\times$ (x, y, z) per hand = 63 dims each, 126 total. Missing hands are zero-filled.
- Face Stream (232 dims) — 60 semantically selected landmarks providing 180 spatial dims, plus 52 MediaPipe blendshape expression scores = 232 total. Full 468-landmark output was reduced to remove noise from irrelevant facial regions.
- Pose Stream (99 dims) — All 33 body landmarks $\times$ (x, y, z) = 99 dims capturing body posture, shoulder orientation, and upper-body context.

The three streams are concatenated per frame to produce a 457-dimensional vector [126 | 232 | 99], yielding a (60 $\times$ 457) input matrix per video. Root-relative coordinate normalization is applied to all streams after extraction to remove position and distance dependency: hand landmarks are centered on the wrist, face landmarks on the nose tip, and pose landmarks on the mid-shoulder point. Extracted features are cached to disk to avoid repeated computation during training.

Data Augmentation

Skeleton-level augmentation was applied stochastically during training (global application probability $p = 0.7$) to increase effective training diversity. Seven stream-appropriate operations were used: spatial rotation ($\pm 5^\circ$ on hand and pose (x, y) pairs), spatial scaling (factor $\in [0.9, 1.1]$), Gaussian noise injection ($\sigma = 0.002$), temporal shifting (up to $\pm 10\%$ frames, $p = 0.3$), time warping (speed $\in [0.8, 1.2]$, $p = 0.25$), random frame dropping (up to 20%, $p = 0.2$), and time masking (contiguous block zeroed, $p = 0.15$). Horizontal flipping was explicitly excluded because left-right handedness

carries semantic meaning in sign language and mirroring can negate sign meaning.

Model Architecture — Multi-Stream Fusion Model

Four architectures were explored iteratively. A baseline Multimodal LSTM (concatenated 457-dim input, BiLSTM with soft attention) achieved $\sim 47\%$ validation accuracy. A Multimodal Transformer (4-layer encoder, $d_{\text{model}}=256$, 8 heads) achieved $\sim 41\%$, requiring more data to converge. A Hybrid LSTM+Transformer reached $\sim 52\%$ by combining local temporal modelling with global attention. The final Multi-Stream Fusion Model (76.3% validation accuracy) separated the three modalities into dedicated sub-networks, producing a 24-percentage-point improvement over the best single-stream baseline.

The Hand TCN Stream uses a three-level Temporal Convolutional Network (TCN) with dilation rates 1, 2, 4 and channel progression $126 \rightarrow 64 \rightarrow 128 \rightarrow 128$; global average pooling produces a 128-dim representation. The Face TCN Stream uses the same TCN structure with wider channels ($232 \rightarrow 128 \rightarrow 256 \rightarrow 256$) to accommodate the richer face feature space, yielding 256 dims. The Pose BiLSTM Stream uses a bidirectional LSTM (hidden=32 per direction, 1 layer) with adaptive average pooling, producing 64 dims; LSTM sequential gating is well-suited to the gradual posture changes during signing[6]. An Attention Fusion Module projects all three streams to a shared 512-dim space, computes per-stream attention scores via a shared two-layer network, normalizes via softmax, and produces a weighted sum followed by LayerNorm and Dropout(0.3). The Classifier Head applies Linear(512 $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(256 $\rightarrow$ 383) to produce logits over all 383 classes. The model has approximately 2.52 million trainable parameters and also returns attention weights per sample, enabling inspection of which modality the model prioritizes for each sign.

Training Configuration

Cross-entropy loss with label smoothing ($\epsilon = 0.1$) was used to prevent overconfidence on the scarce training data. The Adam optimizer was configured with initial learning rate 1×10^{-3} and weight decay 1×10^{-4} . A Cosine Annealing with Warm Restarts schedule ($T_0 = 30$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 1 \times 10^{-6}$) enabled periodic learning rate resets to escape local minima. Batch size was 16. Regularization combined Dropout(0.3) throughout all sub-networks, weight decay, label smoothing, and skeleton augmentation. Early stopping was applied with patience of 20 epochs over a maximum of 120 epochs; the best validation checkpoint was retained. Training was performed on an NVIDIA GeForce MX350 GPU using PyTorch 2.7.1 + CUDA 11.8, MediaPipe 0.10.21, and Python 3.13. Inference runs on CPU for mobile deployment compatibility[8].

Implementation Status

Completed: MediaPipe feature extraction pipeline, root-relative normalization, skeleton augmentation (7 operations), Multi-Stream Fusion Model (TCN + BiLSTM + attention fusion), training with cosine annealing and early stopping, Flask REST API backend (port 5000), React Native mobile integration bridge, real-time inference with Sinhala Unicode rendering via Pillow[8].

D. Sinhala Two-way Communication System for Children with Speaking Disabilities

This component involves designing a Sinhala two-way communication system for children who have difficulty communication (ages 5-10), which is based on Human Computer Interaction (HCI) and Machine Learning (ML). The system allows parents and children to communicate effectively [15], providing information that has been collected through questionnaires.

The overall system consists in three main modules:

1. HCI-based Communication Interface
2. Questionnaire-based Data Input
3. Machine Learning Prediction Module

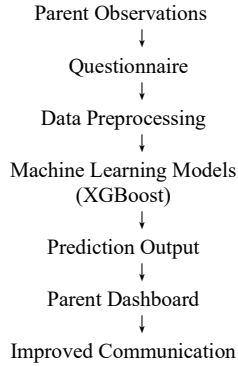


Fig. 3. System Flow Diagram

The system design is based on some key aspects of Human-Computer Interaction, such as usability, accessibility, learnability and low cognitive load.

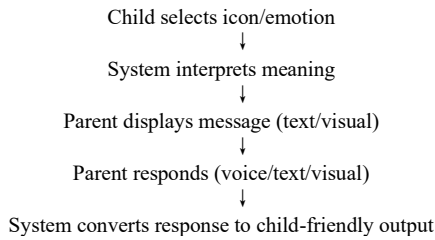


Fig. 4. Interaction Flow

This enables non-verbal two-way communication.

Dataset

The dataset used in this component is collected through a well-structured questionnaire filled by the parents of the children with speaking disabilities ranging from ages 5-10. The questionnaire is designed to capture observable behavioral and communication characteristics of the child in daily life. Each data entry corresponds to one child and includes the following attributes:

- Response to familiar instructions
- Attention and engagement duration
- Ability to express needs and emotions
- Interaction patterns with parents
- Level of independence in daily activities

The outputs for each child correspond to the following aspects of the development process:

- Communication level

- Learning retention
- Attention and engagement
- Independence and confidence
- Instruction understanding

Implementation

XGBoost modeling has been completely integrated into this system. The current state includes questionnaire data preprocessing completed successfully, model training performed for all prediction types identified, comparisons conducted between different types of models (random forest, neural networks), integrating predictions to be displayed in the parent dashboard. The system has the potential to predict based on questionnaire data submitted by the parent.

Expected Outcome

The algorithm is likely to perform very well owing to the fact that it has proved its capabilities in the handling of structured data. Overall Performance, accuracy $\rightarrow$ 80% to 88% and weighted F1 score $\rightarrow$ 78% to 86%. Performance by category, communication level $\rightarrow$ 82% to 88%, instruction understanding $\rightarrow$ 80% to 86%, learning retention $\rightarrow$ 78% to 85%, attention and engagement $\rightarrow$ 79% to 84% and independence ad confidence $\rightarrow$ 80% to 86%.

IV. RESULTS AND DISCUSSION

The performance of the proposed intelligent Sinhala Sign Language communication platform was evaluated by analyzing each component individually, as well as the overall system effectiveness. The integration of multiple assistive modules enables a comprehensive solution that enhances communication, learning, and environmental awareness. A summary of the performance of all four components is presented in Table I.

TABLE I.

Component	Description	Performance
Adaptive Learning System	Personalized SSL learning using reinforcement learning and feedback adaptation	~15% improvement in learning efficiency (estimated)
Environmental Alert Module	CNN-based classification of critical environmental sounds	> 90% classification accuracy
Sign Language Reader	Multimodal recognition using hand, facial, and pose features	76.3% validation accuracy
Two-Way Communication System	ML-based prediction and interactive communication interface	80-88% accuracy

Fig. 5. Summary of four components

The adaptive learning system contributed to improved user engagement and learning outcomes by dynamically adjusting difficulty levels and providing personalized feedback. The use of reinforcement learning allows the system to adapt to individual user performance, resulting in an estimated improvement of approximately 15% in learning efficiency compared to traditional static learning methods. However, further validation through real user studies is required to confirm these findings.

The environmental sound alert module achieved high classification accuracy exceeding 90% across multiple sound categories, including vehicle horns and emergency sirens. The

system operates in near real-time and provides visual alerts that enhance user safety. Despite this, performance can be affected in noisy environments or when multiple sound sources overlap, indicating the need for further improvements in noise robustness and alert prioritization.

The multimodal sign language recognition module demonstrated effective performance, achieving a validation accuracy of 76.3%. By integrating hand gestures, facial expressions, and body posture, the model provides improved recognition compared to single-modality approaches. This multimodal design enables better interpretation of complex and context-dependent signs. However, some challenges remain in distinguishing visually similar gestures, which can be attributed to dataset limitations and variations in user input.

The two-way communication system achieved an overall accuracy between 80% and 88%, effectively supporting interaction between children with speaking disabilities and their caregivers. The interface enables intuitive communication through visual and sign-based outputs, reducing communication barriers in daily use. Nevertheless, the model's performance is influenced by the quality and diversity of the collected dataset, which may impact its generalization across different users.

Overall, the results indicate that the proposed integrated system outperforms conventional isolated solutions by combining multiple assistive functionalities into a unified platform. This integration enhances communication efficiency, learning engagement, and situational awareness. However, certain limitations such as environmental conditions, dataset constraints, and real-time processing challenges must be addressed in future work to improve system robustness and scalability.

V. CONCLUSION

This research presented an intelligent Sinhala Sign Language communication platform designed to address key challenges faced by individuals with hearing and speech impairments. By integrating multiple assistive functionalities into a single system, the proposed solution provides a comprehensive approach to improving communication, learning, and environmental awareness. The platform combines multimodal sign language recognition, adaptive learning, environmental sound detection, and an interactive two-way communication interface to support more effective and inclusive user interaction.

The experimental results demonstrate that the system achieves reliable performance across all components, with strong accuracy in sign recognition, sound classification, and communication prediction. The use of multimodal inputs enhances recognition capability, while adaptive learning techniques improve user engagement and learning efficiency. Additionally, the incorporation of environmental awareness and bidirectional communication features contributes to a more practical and user-centered solution for real-world applications.

Despite these contributions, certain limitations remain, including dependency on dataset quality, sensitivity to environmental conditions such as noise and lighting, and the need for further real-world validation. Addressing these challenges will be essential to improving system robustness and scalability.

Future work will focus on expanding datasets, optimizing models for on-device deployment, and conducting user-centered evaluations to enhance usability and performance. Overall, this study highlights the potential of combining artificial intelligence with human-computer interaction principles to develop effective and accessible assistive technologies, contributing to more inclusive communication solutions in the context of Sinhala Sign Language.

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